

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JNANA SANGAMA”, BELAGAVI, KARNATAKA-590018



A Report on

“Privacy Coins and Anonymous Transaction”

Submitted By

K.N.MEGHANA 4MN22CS020

L.SANJANA 4MN22CS021

*Submitted in partial fulfillment of the requirement for the award of the degree of
Bachelor of Engineering in Computer Science & Engineering*

Under the guidance of

Prof. Madhu.B

Assistant Professor

Department of Computer Science & Engineering

MIT Thandavapura



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAHARAJA INSTITUTE OF TECHNOLOGY THANDAVAPURA

Just Off NH 766, Nanjanagudu Taluk, Mysore District – 571302

(Approved by AICTE, Accredited by NBA, New Delhi and Affiliated by VTU, Belagavi)

2024-2025

ABSTRACT

Privacy coins are cryptocurrencies designed to offer secure, untraceable, and anonymous financial transactions on blockchain networks. Unlike traditional cryptocurrencies such as Bitcoin, where transaction details are publicly visible, privacy coins use advanced cryptographic techniques to hide the sender's and receiver's identities and the transaction amount. While these coins support legitimate needs for privacy and security, they have also raised concerns due to their potential misuse in illegal activities.

CONTENTS

SL. No.			Index	Page No.
1			INTRODUCTION	1
	1.1		Privacy Coins	1
		1.1.1	Key Features	1
		1.1.2	Techniques for Privacy in Block Chain	1-2
		1.1.3	Examples	2
	1.2		Anonymous Transactions	2
		1.2.1	Truly anonymous Transactions	2
		1.2.2	Techniques for anonymous transactions	3
		1.2.3	Examples	3
		1.2.4	Limitations of Anonymous Transactions	4
		1.2.5	Real world use case of anonymous transaction	4
			Snapshots	5

1. INTRODUCTION

Privacy coins are a type of cryptocurrency designed to keep transaction details hidden from the public. Unlike regular cryptocurrencies like Bitcoin, where all transactions are visible on the blockchain, privacy coins use special technologies to hide the sender, receiver, and amount. Examples include Monero and Zcash, which use methods like ring signatures and zero-knowledge proofs. These coins help protect user privacy but also raise concerns about illegal use. As digital finance grows, privacy coins play an important role in offering secure and anonymous transactions.

1.1 PRIVACY COINS

Privacy coins are a special category of cryptocurrencies that prioritize user anonymity and transaction privacy. Unlike typical cryptocurrencies such as Bitcoin or Ethereum, which have transparent public ledgers, privacy coins obfuscate key details of transactions—such as the sender, receiver, and transaction amount.

1.1.1 Key Characteristics:

- Confidentiality by design
- Built-in obfuscation techniques
- Use advanced cryptographic methods

1.1.2 Techniques for Privacy in Blockchain

1. Mixing Protocols

- Tools like CoinJoin combine multiple users' transactions to confuse chain analysis.
- Used in wallets like Wasabi and Samourai.

2. Zero-Knowledge Proofs (zk-SNARKs/zk-STARKs)

- Allow verification without revealing any data.
- Used by Zcash and increasingly in Layer 2 Ethereum solutions.

3. Stealth Addresses

- Generate one-time addresses for each transaction.
- Used by Monero to hide recipient details.

4. Ring Signatures

- Multiple possible signers are included, hiding the true one.

- Makes sender indistinguishable.

5.Mimblewimble

- Obscures all transaction data and allows transactions to be aggregated and compressed.
- Lightweight and scalable (e.g., Beam, Grin).

6.Layer 2 Privacy Protocols

- Protocols like Aztec and zkSync (Ethereum Layer 2) add privacy to otherwise public smart contracts.

1.1.3 Examples:

Coin	Technology Used	Privacy Feature
Monero (XMR)	Ring signatures, stealth addresses, RingCT	Default full anonymity
Zcash (ZEC)	zk-SNARKs (zero-knowledge proofs)	Optional shielded transactions
Dash	Coin mixing (PrivateSend)	Optional privacy
Beam & Grin	Mimblewimble protocol	Lightweight, scalable privacy

1.2 ANONYMOUS TRANSACTIONS

Anonymous transactions are those where the identities of the sender and receiver, as well as transaction amounts, are concealed or obfuscated from public view.

Unlike traditional cryptocurrencies like Bitcoin, where every transaction is recorded transparently on the blockchain, anonymous transactions aim to protect user privacy and transaction confidentiality.

1.2.1 Truly Anonymous Transactions:

- Hide sender, receiver, and amount.
- Use advanced cryptographic protocols.
- Cannot be easily traced, even with advanced tools.

1.2.2 Techniques for anonymous transactions

Technique	Description	Purpose
Coin Mixing / Tumbling	Combines coins from multiple users, mixes them, then redistributes	Breaks transaction trails
CoinJoin	Combines multiple transactions into one shared transaction	Hides sender-receiver linkage
Stealth Addresses	Generates unique addresses for every transaction	Hides recipient identity
Ring Signatures	Blends the sender's input with others	Conceals sender identity
Confidential Transactions (CTs)	Encrypts the transaction amount	Hides value transferred
zk-SNARKs / zk-STARKs	Zero-knowledge proofs verify transactions without revealing data	Enables complete anonymity
Mimblewimble	Eliminates addresses, encrypts amounts, merges transactions	Scalable privacy solution
Dandelion++	Obfuscates how transactions propagate through the network	Hides the user's IP address

1.2.3 Examples

Blockchain / Coin	Privacy Features	Level of Anonymity
Monero (XMR)	RingCT, stealth addresses, Dandelion++	Full
Zcash (ZEC)	zk-SNARKs (optional privacy)	High (if shielded)
Grin / Beam	Mimblewimble protocol	High
Secret Network	Encrypted smart contract state	Medium to High
Tornado Cash (Ethereum Mixer)	zk-SNARK-based mixing for ETH and ERC-20	High, but banned in some countries

1.2.4 Limitations of Anonymous Transactions

Limitation	Explanation
Performance Overhead	Cryptographic privacy methods (e.g., zk-SNARKs) are computationally expensive.
Complexity	Harder to use for non-technical users; risk of misconfiguration.
Size & Speed	Privacy transactions can be larger and slower to process.
Auditability	Makes regulatory compliance and forensic analysis difficult.
Optional Privacy Can Be Misused	If not default (like in Zcash), users may not use privacy features properly.

1.2.5 Real-World Use Cases for Anonymous Blockchain Transactions

1. Personal Financial Privacy

E.g., People who don't want their employer, government, or competitors to track their expenses.

2. Political Dissidents & Activists

Operating in authoritarian regimes where financial activity can lead to persecution.

3. Whistleblowers

Anonymous donations or tips without risk of exposure.

4. Private Business Transactions

Companies seeking to protect supply chain or pricing strategies.

SNAPSHOTS



